# Introduction to Algorithms

| $n$ | $n^5$ | $2^n$ | $3^n$ | Complexity / Question |
|---|---|---|---|---|
| 0.00006 sec. | 13 min. | 366 cent. | $1.3 \times 10^{13}$ cent. | For 1,000,000 op per second computer, N=60, how much time needed to compute ? |
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| $18 \times 10^9$ | 112 | 34 | 21 | For same computer, 5 CPU hours, which input size can be reached? |
| $18 \times 10^{10}$ | 178 | 37 | 23 | For 10 times faster computer, 5 CPU hours, which input size can be reached? |

# Introduction to Algorithms

Definition:

An algorithm is called *tractable* if its complexity is $O(n^k)$, where n is the input size and k is a constant.

What about $O(n \log n)$ algorithm?

Definition:

A problem is called *tractable* if it is solvable by a tractable algorithm.

What if we have exp. time algorithm?